Blame as Sensemaking: A Computational Mapping of Target Framings Across Online AI Discourse

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Abstract

The increased human-likeness of AI agents has intensified discussions on the perceived moral agency of AI, yet insufficient attention has been paid to the everyday language used to describe its actions. Addressing this gap, this study computationally analyzes anthropomorphic blame language in AI-related discussions on Reddit. Focusing on blame situations in the two largest AI-related subreddits, we explore the semantic similarities across blame language targeting AI, human, organizational, and societal actors. Analysis of post-level embeddings reveals high overall semantic similarity across targets, with distinctive topical profiles. Blame directed at AI agents aligns closest to society-level blaming across topical anchors, while moral action anchors reveal structures unique to AI blaming. These findings suggest that AI is recognized as blameworthy, but the tone and style of its linguistic description are largely shared with other actors. We highlight the implications of these patterns in the public sensemaking of responsible AI design and governance.

CCS Concepts

• **Human-centered computing → Collaborative and social computing**; **Empirical studies in HCI**; • **Computing methodologies → Natural language processing**.

Keywords

Blame, Public sensemaking, Online discourse, Responsibility, Moral agency, Human-AI Interaction

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1 Introduction

Artificial Intelligence (AI) technologies are being increasingly developed to assist and emulate a wider range of human abilities. As these systems become embedded in everyday interactions, people make sense of their actions by relying on human-centered metaphors and newly coined terms such as "thinking", "creating", or "hallucinating" [8, 25]. While such linguistic conventions do not always reflect explicit beliefs about AI agency, they shape narratives in which AI systems are recognized as intentional actors [27, 29]. Notably, in online conversations, AI is blamed for lying, misleading users, or causing harm [9] despite the absence of a shared belief that AI possesses moral agency [22, 28]. In mass-oriented communication environments, such as social media, the circulation of these narratives can actively reconfigure how realities about AI are collectively understood [33].

Prior work in human-computer interaction (HCI) has recognized the risks of anthropomorphism [1, 2], yet AI systems development fosters human-likeness for user satisfaction and engagement [16, 23, 36]. Under these circumstances, questions arise on how perceived moral agency reshapes discussions of responsibility and blame in human-AI interactions.

Online discourse offers a critical site for examining how moral agency is prescribed through blame and how sociotechnical imaginaries of AI responsibility are shaped [13, 15]. While existing work on moral agency and blame in AI has focused on belief-based judgments rooted in individual knowledge and ethical reasoning [9, 14, 19, 21, 37], less attention has been paid to how responsibility is constructed through surface-level language in organic public discourse.

Using this gap as an opportunity, the present study examines blaming patterns of AI-related targets in online discourse. We define blame as the recognition of a target's action, and design a concrete text classification pipeline to capture the definition at scale. By conducting embedding analyses on AI-related Reddit posts, this study investigates how blame is linguistically structured across different entities involved in AI systems.

This study highlights three contributions. First, it proposes a way to operationalize blame as an emergent act of sensemaking rather than a well-deliberated action by allowing both the situation and the target of blame to be fluid. Second, it examines the language enacted by AI and human entities without presupposing fixed boundaries of moral standing, focusing on how different targets are *spoken of* rather than how they are *thought of*. Third, it introduces a scalable method of analysis to capture anthropomorphized linguistic tendencies in the most up-to-date discussions on AI by leveraging a platform where fast-paced developments are actively reflected.

Taken together, the study addresses the following research questions:

- RQ1: How do blame topics and linguistic structures vary across AI-related targets in social media discourse?

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- RQ2: To what extent is blame directed at AI linguistically distinct from blame directed at other entities, namely the company, the developer, or the social structure behind it?

2 Related Work

2.1 Blame as Belief, Blame as Practice

Prior work on blame attribution has largely centered on the attachment between observable actions and underlying intentions when assigning responsibility [5, 17, 24, 32]. However, a growing body of empirical studies has contributed to the recognition of blame as an action that can be isolated from the belief or normative knowledge on the applicability of moral agency [4, 14, 31, 37]. Hong et al. [14] show that blame assignment in human-AI interactions is driven more by expectancy violation than by perceived responsibility of the actant. Similarly, Zhang et al. [37] demonstrate that in moral dilemma situations, whether an action occurred matters more than who performed it.

These findings suggest that everyday blaming practices are structured by evaluations of action and outcome rather than by ontological judgments about agential legitimacy. However, much of this work conceptualizes blame as an internal judgment elicited through experimental settings, which offers limited insight into how blame is expressed, circulated, and stabilized in public discourse. This gap motivates an examination of blame as a situated linguistic practice through which responsibility is publicly constituted.

2.2 Anthropomorphic Language and Concerns

A growing body of research has examined how anthropomorphic language is used to describe non-human entities, particularly AI systems. Prior work has shown that surface-level linguistic patterns in public discourse serve as a proxy for capturing how moral agency and responsibility are socially constructed around AI [26, 29].

Recent works by Cheng et al. [3] and DeVrio et al. [7] contributed to clarifying the definition of anthropomorphic language through computational classification and qualitative taxonomy-building approaches. While Cheng et al. examine anthropomorphic language used about non-human entities, DeVrio et al. focus on human-like language produced by LLMs, highlighting its social role as a communicator.

Building on a social media analysis, Heaton et al. [11] identify a potential responsibility gap in Twitter discourse, where AI systems are simultaneously attributed agency and passivity. Similarly, Franklin et al. [9] demonstrate that blame attribution patterns previously observed in experimental settings can be replicated at scale in naturalistic online data. Responding to Franklin et al.'s call for future work, we extend this line of inquiry by applying more recent embedding-based models to examine how blame is framed and linguistically structured in contemporary AI discourse.

3 Methods

3.1 Data

Reddit posts from two representative AI discussion subreddits, *r/artificial* and *r/ArtificialInteligence*, were extracted via 'Project Arctic Shift' [12]. Posts from January 1, 2023, to December 31, 2024, were collected to focus on discourse surrounding generative models, as ChatGPT was launched at the end of 2022. The preprocessing steps included removing posts shorter than 15 characters, non-English posts, duplicates, hyperlinks, and markers of deletion. As a result, the final dataset consisted of 81,871 posts.

3.2 Text Classification

Text classification was conducted in three steps: *negative reaction detection*, *target specification*, and *blame detection*. Based on a strictly defined annotation guideline, a human annotator classified a total of 424 posts for negative evaluation, 112 posts for target specification, and 92 posts for blame detection. Negative reaction detection was performed using an NLI-based DeBERTa zero-shot classifier [18], while the two subsequent steps utilized the GPT 4o mini API with few-shot prompting. The pipeline yielded a final set of 3,752 posts. F1 scores and accuracy across all steps indicated sufficient reliability for downstream analyses (see Table 2).

3.2.1 *Negative Reaction Detection.*

Negative reaction detection was aimed at eliminating non-negative and purely objective content. Appraisal theory [34] and the annotation guidelines for subjectivity classification by Wiebe et al. [35] were used for the conceptualization of this process. To improve annotator consistency, the list of negative emotions from the GoEmotions dataset [6] was used as a practical reference. A classification threshold of 0.4 was applied to retain borderline cases, as the subsequent blame detection stage nests negativity as a premise. The zero-shot hypothesis and label options can be found in the Supplementary material.

3.2.2 *Target Specification and Blame Detection.*

The target specification and the blame detection followed a concrete conceptual framework that identified blame attribution without invoking normative evaluations of moral agency. In this research, *Target* was defined as the object toward which the author of a post negatively reacts. A total of 7 target options were specified for each post: the AI model; companies developing AI models; companies deploying the systems; human developers behind the company; the broader social structures; individual users; and others (see Supplementary material). *Blame* was operationalized as the recognition of a target's action as a causal source of the negative response (e.g., action verbs such as *lie* or *steal* attributed to the target). These two stages were conducted simultaneously for both the human annotation and LLM-based classification processes.

3.3 Embedding-based Semantic Analyses

3.3.1 *Topic Modeling.*

Topical variations of blame were explored by conducting BERTopic modeling [10] separately on target-specific corpora using document-level embeddings from a pretrained sentence-transformer model [30]. While seven target options were identified during classification, our analysis focused on four dominant targets – AI model, AI company, AI developer, and Society — as this study centers on developer and system-level responsibility rather than individual user responsibility. Posts implicating multiple targets were included in each corpus. Topic outputs were further grouped into super-clusters to capture higher-level framing structures across targets.

3.3.2 *Cluster Comparisons and 'One target vs. Rest' Scores.*

To compare stylistic differences in blame across targets, post embeddings

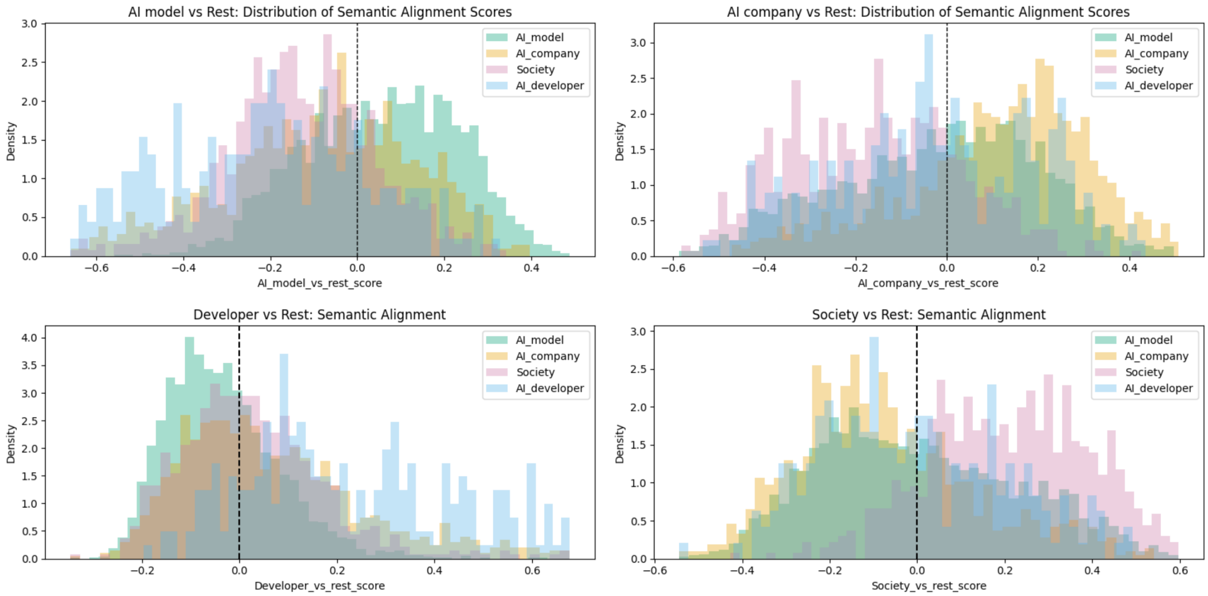

Figure 1: 'One target vs. Rest' alignment score distributions across targets. Partial non-overlap among histograms indicates target-specific semantic differentiation in the embedding space. Posts within each target group are expected to exhibit positive skew relative to their corresponding centroid, though the magnitude of differentiation varies across targets.

were compared using cosine similarity between target centroids. In addition, 'One target vs. Rest' scores were computed to assess the distinct characteristics of each target. The vector equation for the scores is denoted as follows.

$$\mathbf{v}_t = \mathbf{c}_t - \frac{1}{|\mathcal{T}| - 1} \sum_{k \neq t} \mathbf{c}_k$$

where $\mathbf{c}_t$ denotes the centroid embedding of posts blaming target t, and $\mathcal{T}$ denotes the set of analyzed targets. Post-level alignment scores were obtained by computing the cosine similarity between each post embedding and the corresponding contrast vector. Permutation-based tests were conducted to evaluate the statistical reliability of observed means and distributional differences.

Posts blaming AI exclusively were further filtered using topical and moral action anchors. Among 3,029 AI-targeting posts, 1,717 were identified as AI-only blame instances. Anchor words were selected inductively through qualitative inspection of frequently occurring terms identified during annotation and BERTopic analysis.

4 Findings

4.1 Finding 1: The four targets largely share a common semantic structure of blame, although each target still exhibits distinct alignment patterns.

The four target clusters exhibit high cosine similarity of centroids, with values ranging from 0.91 to 0.96 across all six pairs (see Table 3). Moreover, AI-targeted blaming posts substantially overlap with those of the other three targets. This is partly explained by the corpus size, where 3,029 posts were assigned for the AI model, 993 for the company, 231 for the developer, and 662 for society. Thus, the semantic distinctions are not visibly pronounced through multi-targeted centroid comparison.

To account for this problem, target-specific blaming structures were further examined using 'One target vs. Rest' scores. AI models and AI companies exhibit relatively low levels of semantic distinctiveness, whereas AI developers and society display more pronounced distinctive features. This pattern is reflected in larger mean differences from the second-closest target alignment and in distributional regions with minimal overlap across targets (Figure 1; Table 1). Due to non-independence across targets arising from multi-labeling, statistical significance was assessed via permutation testing (5000 iterations). Both mean separation (T_{mean}; variance of mean alignment scores) and distributional divergence (T_{shape}; mean pairwise Wasserstein distance) were statistically significant across all targets (permutation p=0.0002 for both permutation tests; see Table 4).

4.2 Finding 2: Topical divergence coexists with converged semantic structures of blame.

Although the semantic styles through which blame is articulated remain largely similar across targets (Finding 1), the super-topic clustering reveals that distinct targets are criticized through shared semantic and evaluative structures across seemingly unrelated topics (see Table 5). For instance, a topic represented by the terms [ask, tell, clear, output] from the AI model corpus was grouped with [case, legal, law, chatbot] from the AI company corpus.

Notably, the largest super-topic encompassed all four targets, with the same terms co-occurring with a different combination of words. The term 'AGI' appeared both in the company and the

Table 1: 'One target vs. Rest' score alignment mean across targets. Rows correspond to focal target groups, and columns report mean alignment scores with each target centroid. Bold values indicate the dominant alignment within each row. The final column shows the mean gap between the highest and second-highest alignment scores, capturing the relative distinctiveness of each group.

Target Group	Model vs. rest	Company vs. rest	Developer vs. rest	Society vs. rest	Mean diff. from 2nd highest
AI Model	**0.054**	-0.018	-0.036	0.001	0.053
AI Company	-0.083	**0.113**	0.066	-0.076	0.047
AI Developer	-0.202	-0.026	**0.225**	-0.011	0.236
Society	-0.130	-0.162	0.024	**0.209**	0.185

society corpora, but were each grouped with [human, intelligence, humanity], proposing more existential concerns, and [think, years, government], representing a more pragmatic questioning of future governance.

These results indicate that while targets give rise to different topical frames of criticism, the linguistic style through which blame is organized converges across targets.

4.3 Finding 3: AI-only blame frequently aligns better with society-level blame for topical anchors, while unique structures arise for moral action verbs.

When AI is the sole target of blame, the semantic structures of blame for the majority of topics align better with system-blaming frames rather than with their own, except for several cases with moral action verbs.

The distinctive semantic patterns of AI models were further examined by navigating different anchor terms within AI-only blaming posts. With topical anchors such as humanity, jobs, safety, and privacy, a larger portion of AI-only blaming posts aligned more closely with society-blaming language. The cases of "error" and "bias" were notable exceptions, indicating that the posts involving those anchors were presented in a structure unique to AI models.

For moral action verbs, the alignment with society-blaming persisted, but a higher proportion of semantic characteristics unique to AI model blaming was discovered. Posts involving the verbs "fail," "lie," "ignore," and "steal" had a higher percentage of alignment with AI model blaming language rather than with Society-blaming (Figure 2). This indicates that although the same verbs are used to describe different human and non-human actions, the intended uses may be distinct from one another.

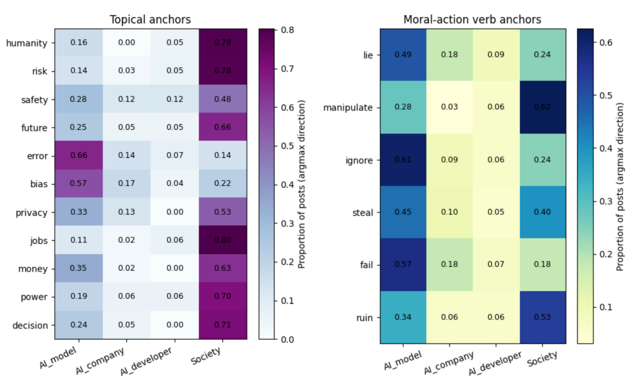

Figure 2: Target alignment proportions of AI-only blaming posts by anchor words. Anchor terms were identified through exploratory testing of high-frequency vocabulary within blame posts and retained if they occurred in at least 15 posts. Cells indicate the proportion of posts containing each anchor term that align best with the corresponding target centroid.

5 Discussion

Our findings suggest that in Reddit discourse, AI is recognized as an agent capable of practicing moral actions, with the same human verbs used in its own semantic meaning. Although this finding does not directly translate into the attribution of moral agency to AI systems, it highlights the current state of collective sensemaking, as expressed through public testimony. The use of shared grammar across targets, especially encompassing AI systems, raises the question of whether the responsibility gap can widen as the public internally processes and outwardly stabilizes a narrative that AI systems have their own problems at the same categorical level as human problems. That is, the different moral positions and agendas assigned to each target may be blurred under a shared sensemaking logic.

The strong semantic alignment between AI-only blame and society-level blame can be interpreted in multiple ways. One possible explanation is that the public does not primarily perceive AI as an individual agent with a coherent self, but rather as a broader technological system. This interpretation is also supported by our classification process, where general expressions referring to AI technologies were included under 'AI model'.

At the same time, this alignment may reflect public uncertainty regarding who should be held responsible for AI-related outcomes. Society-level blame often emerges in situations where responsibility cannot be clearly attributed to a single actor [20]. In the case of AI, the rapid development cycle may leave the public amid a novelty effect, where mindless anthropomorphization outweighs shared understandings that have yet to be established. In this sense, depending on how AI technologies and their public imaginaries are shaped through public discourse, responsibility risks being diffused toward implicit societal actors rather than prescribed explicitly to accountable entities. Therefore, our findings invite relevant stakeholders, such as systems designers and policymakers, to reflect on how normative guidelines could better support clearer responsibility attribution within the evolving AI discourse. In doing so, we highlight the role of social media as the main platform on which new realities are created, and the value of connecting with the public through such sources.

6 Limitations

Although the embedding-based analysis offers a broad landscape of how blame is positioned similarly or differently across targets, a comprehensive examination of variations at finer analytical levels remains underexplored. For the text classification pipeline, the reliability of the ground truth should be strengthened by incorporating multiple annotators. In addition, further justification could be achieved by systematically testing alternative classification and embedding models. Finally, as this study relies on Reddit data, which predominantly reflects North American and English-speaking user populations, the findings should be interpreted with caution when considering global generalizability.

7 Conclusion and Future Work

This study examined surface-level expressions of blame directed at AI to explore emerging possibilities for human-AI relationships constructed through public discourse. The findings point to an unresolved ambiguity regarding the expected responsibility in AI-driven problem situations. As the work continues, further qualitative assessments of embedding-based results will be conducted to interpret how different topics converge into shared semantic structures. Future work may also apply narrower filters for specific types of AI to examine differences in anthropomorphization across technologies. Additional actor categories, such as individual users, could also be incorporated into the analysis to examine how blame diffuses and coalesces across multiple targets. Moreover, a study on how these linguistic tendencies actually restructure people's beliefs about responsible AI could be a valuable addition to the discussion.

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A Appendix: Extended Tables

A.1 F1 Scores and Accuracy Table for Classification Stages

Classification Stage	F1 Score (Weighted)	Accuracy
Negative Reaction	0.836	0.611
Target: AI Model	0.809	0.812
Target: AI Company	0.830	0.820
Target: Other Company	0.894	0.891
Target: AI Developer	0.925	0.919
Target: Society	0.930	0.929
Target: Users	0.960	0.950
Target: None	0.916	0.925
Blame	0.829	0.829

Table 2: Weighted F1 and accuracy were used to validate model performance against human annotations.

A.2 Cosine Similarity of Target Cluster Centroids

	AI Model	AI Company	AI Developer	Society
AI Model (n=3,029)	1.000			
AI Company (n=993)	0.955	1.000		
AI Developer (n=231)	0.917	0.955	1.000	
Society (n=662)	0.931	0.905	0.928	1.000

Table 3: Cosine similarity measuring directional alignment between target embedding centroids.

A.3 Permutation Test Results for 'One target vs. Rest' Score Mean and Distribution

Comparison	Test Statistic	Observed	p-value
AI Model vs. Rest	T_{mean}	0.00870592	0.0002
	T_{shape}	0.139593	0.0002
AI Company vs. Rest	T_{mean}	0.00946152	0.0002
	T_{shape}	0.139864	0.0002
AI Developer vs. Rest	T_{mean}	0.00931750	0.0002
	T_{shape}	0.137338	0.0002
Society vs. Rest	T_{mean}	0.01141130	0.0002
	T_{shape}	0.145601	0.0002

Table 4: Permutation tests (5000 iterations) were conducted to evaluate target distinctions under multi-label dependencies. For each permutation, two statistics were computed: (1) T_{mean}, the variance of mean alignment scores across targets (capturing differences in central tendency), and (2) T_{shape}, the mean pairwise Wasserstein distance between target distributions (capturing distributional separation). p-values reflect the proportion of permuted statistics greater than or equal to the observed value.

A.4 Super-Clustered BERTopic Outputs

Super-Topic	Target	Total Docs	Topics
1	AI Model	50	[response, trying, asked, did], [ask, tell, clear, output]
	AI Company	16	[case, legal, law, chatbot]
	AI Developer	0	–
	Society	22	[sub, link, posts, post]
2	AI Model	69	[said, story, image, users]
	AI Company	25	[search, content, results, website]
	AI Developer	–	–
	Society	60	[human, risks, safety, nuclear], [robot, videos, online, targets]
3	AI Model	146	[generated, text, using, knowledge], [story, write, answer, questions]
	AI Company	66	[altman, am, ceo, sam altman], [content, search, tool, writing]
	AI Developer	–	–
	Society	23	[humans, human, intelligence, problems]
4	AI Model	415	[tried, best, free, ve], [stop, end, humans, ask], [story, asked, knows, video], [using, ve, tried, working], [answer, problems, questions, read], [open, code, doesn, asked], [source, major, new, open], [generation, users, real, including], [money, software, years, year], [years, learn, learning, come]
	AI Company	274	[agi, human, intelligence, humanity], [vision, 4o, image, really], [course, step, write, user], [image, images, content, provide], [billion, 100, idea, failed], [power, users, resources, capable]
	AI Developer	162	[data, use, information, users], [open, human, open source, new], [board, ceo, employees, letter], [musk, elon, elon musk, twitter], [god, tower, reason, led]
	Society	182	[jobs, job, going, robots], [board, employees, letter, source], [law, data, image, federal], [eu, china, act, source], [agi, think, years, government], [course, students, knowledge, using], [projects, big tech, tech, study]
5	AI Model	58	[media, generated, tools, social], [media, post, generated, social]
	AI Company	40	[results, test, user, simple]
	AI Developer	–	–
	Society	18	[false, eu, data, generated]

Table 5: Super-topics were identified by re-grouping topic embeddings (extracted per corpus) using HDBSCAN, requiring a minimum of three total topics and two neighboring topics per cluster. As BERTopic outputs centroids of post clusters, the resulting super-topics reflect semantic similarities across targets.